

ADAR'S STUDY SERIES

LINEAR ALGEBRA

PART 1 · FOUNDATIONS

Systems of Linear Equations · Augmented Matrices · Row Reduction · Echelon Form

Deep Violet & Amber Edition

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1 • INTRODUCTION

Linear Algebra is the branch of mathematics that studies linear equations, systems of linear equations, vectors, matrices, and linear transformations.

Main Goal

- Find the values of unknown variables that satisfy one or more linear equations simultaneously.

Applications

- Artificial Intelligence (AI)
- Machine Learning (ML)
- Data Science
- Computer Graphics
- Computer Vision
- Signal Processing
- Engineering

2 · SYSTEM OF LINEAR EQUATIONS

A system of linear equations is a collection of two or more linear equations involving the same variables.

General Form

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m \end{aligned}$$

3 · WORKED EXAMPLE — TWO VARIABLES

Find x and y such that:

$$\begin{aligned} x - y &= -2 \\ 3x - 6y &= -15 \end{aligned}$$

Method 1 — Substitution

From equation (1):

$$\begin{aligned} x - y &= -2 \\ \Rightarrow x &= y - 2 \end{aligned}$$

Substitute into equation (2):

$$\begin{aligned} 3(y - 2) - 6y &= -15 \\ \Rightarrow 3y - 6 - 6y &= -15 \\ \Rightarrow -3y &= -9 \\ \Rightarrow y &= 3 \end{aligned}$$

Then:

$$x = y - 2 = 3 - 2 = 1$$

$$\text{Answer: } (x, y) = (1, 3)$$

Limitation: the substitution method becomes messy as the number of variables and equations grows.

Method 2 — Elimination

Multiply equation (1) by 3:

$$3x - 3y = -6$$

Subtract from equation (2):

$$\begin{aligned} (3x - 6y) - (3x - 3y) &= -15 - (-6) \\ \Rightarrow -3y &= -9 \\ \Rightarrow y &= 3 \end{aligned}$$

Substitute back into equation (1):

$$\begin{aligned} x - 3 &= -2 \\ \Rightarrow x &= 1 \end{aligned}$$

$$\text{Answer: } (x, y) = (1, 3)$$

Geometric Interpretation

Each equation represents a straight line in the 2D plane.

- Intersecting lines $\rightarrow$ One unique solution
- Parallel lines $\rightarrow$ No solution
- Same line $\rightarrow$ Infinitely many solutions

For this example, the two lines intersect at exactly one point:

$$\text{Intersection Point: } (1, 3)$$

4 · WORKED EXAMPLE — THREE VARIABLES

Find x , y and z such that:

$$x - 2y + 4z = 1$$

$$x + y + 7z = 3$$

Solving the System

Subtract equation (1) from equation (2):

$$(x + y + 7z) - (x - 2y + 4z) = 3 - 1$$

$$\Rightarrow 3y + 3z = 2$$

$$\Rightarrow y = \frac{2}{3} - z$$

Substitute into equation (1):

$$x - 2\left(\frac{2}{3} - z\right) + 4z = 1$$

$$\Rightarrow x = 1 + \frac{4}{3} - 2z - 4z$$

$$\Rightarrow x = \frac{7}{3} - 6z$$

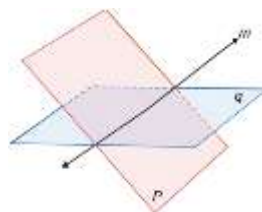
Let the free variable $z = t$ (t is a parameter):

$$x = \frac{7}{3} - 6t, \quad y = \frac{2}{3} - t \quad \text{and} \quad z = t \quad (t \in \mathbb{R})$$

Geometric Interpretation

Each equation represents a plane in 3-dimensional space. Two (non-parallel) planes generally intersect in a straight line — not a single point.

Therefore, this system has infinitely many solutions, forming a line parameterised by t .



5 · LINEAR EQUATIONS IN n VARIABLES

General Form

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$

Where:

- $x_1, x_2, \dots, x_n \rightarrow$ Variables
- $a_1, a_2, \dots, a_n \rightarrow$ Coefficients
- $b \rightarrow$ Constant

General Linear System

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m \end{aligned}$$

6 • THE AUGMENTED MATRIX

The augmented matrix packages every coefficient and constant of a linear system into a single array, separating the coefficients from the constants with a vertical bar.

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} & | & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & | & b_2 \\ : & : & & : & | & : \\ a_{m1} & a_{m2} & \dots & a_{mn} & | & b_m \end{bmatrix}$$

- Left part → Coefficient Matrix
- Right part → Constant Column

Goal of Row Reduction

Reduce the augmented matrix using elementary row operations until the solution of the system can be read off directly.

7 · ELEMENTARY ROW OPERATIONS (EROs)

1. Row Scaling

$$R_i \rightarrow kR_i \quad (k \neq 0)$$

Multiply every entry of a row by a non-zero constant.

2. Row Replacement

$$R_j \rightarrow R_j \pm kR_i$$

Add/subtract a multiple of one row to another row.

3. Row Interchange

$$R_i \leftrightarrow R_j$$

Swap the positions of two rows.

Key Fact: Elementary Row Operations never change the solution set of the system — they only change its appearance.

8 · ROW REDUCTION — EXAMPLES REVISITED

Example 1 (two variables) — via Augmented Matrix

Starting Augmented Matrix

$$\begin{bmatrix} 1 & -1 & | & -2 \\ 3 & -6 & | & -15 \end{bmatrix}$$

Apply the operation:

$$-3R_1 + R_2 \rightarrow R_2$$

Result

$$\begin{bmatrix} 1 & -1 & | & -2 \\ 0 & -3 & | & -9 \end{bmatrix}$$

From Row 2:

$$-3y = -9 \Rightarrow y = 3$$

From Row 1:

$$x - y = -2 \Rightarrow x = 1$$

Answer: $(x, y) = (1, 3)$

Previous Example 2 (three variables) — via Augmented Matrix

Starting Augmented Matrix

$$\begin{bmatrix} 1 & -2 & 4 & | & 1 \\ 1 & 1 & 7 & | & 3 \end{bmatrix}$$

Apply the operation:

$$R2 - R1 \rightarrow R2$$

Result

$$\begin{bmatrix} 1 & -2 & 4 & | & 1 \\ 0 & 3 & 3 & | & 2 \end{bmatrix}$$

From Row 2:

$$3y + 3z = 2 \quad \Rightarrow \quad y = \frac{2}{3} - z$$

Substitute the expression of y obtained from Row 2 into Row 1:

$$x = \frac{7}{3} - 6z$$

Let $z = t$ (free parameter):

$$x = \frac{7}{3} - 6t, \quad y = \frac{2}{3} - t \quad \text{and} \quad z = t \quad (t \in \mathbb{R})$$

9 · ECHELON FORM — OVERVIEW

Learning Goals

- Learn Echelon Form (EF) and Reduced Row Echelon Form (RREF).
- Solve linear systems using systematic Row Reduction.
- Determine the number of solutions of a linear system.

Types of Solutions

Inconsistent System

- No Solution

Consistent System

- One Unique Solution
- Infinitely Many Solutions

10 · KEY POINTS & SUMMARY

- Linear Algebra studies linear equations and systems of linear equations.
- An augmented matrix packages coefficients and constants into one array.
- Row reduction simplifies a system without changing its solution set.
- Three Elementary Row Operations: Row Scaling, Row Replacement, Row Interchange.
- Echelon Form (EF) and Reduced Row Echelon Form (RREF) are standard forms used to solve systems efficiently.
- A linear system may have: No Solution, One Unique Solution, or Infinitely Many Solutions.

END OF PART 1 · NEXT: PART 2 — ECHELON FORM IN DEPTH